

AI-driven Dispensing of Coral Reseeding Devices for Broad-scale Restoration of the Great Barrier Reef

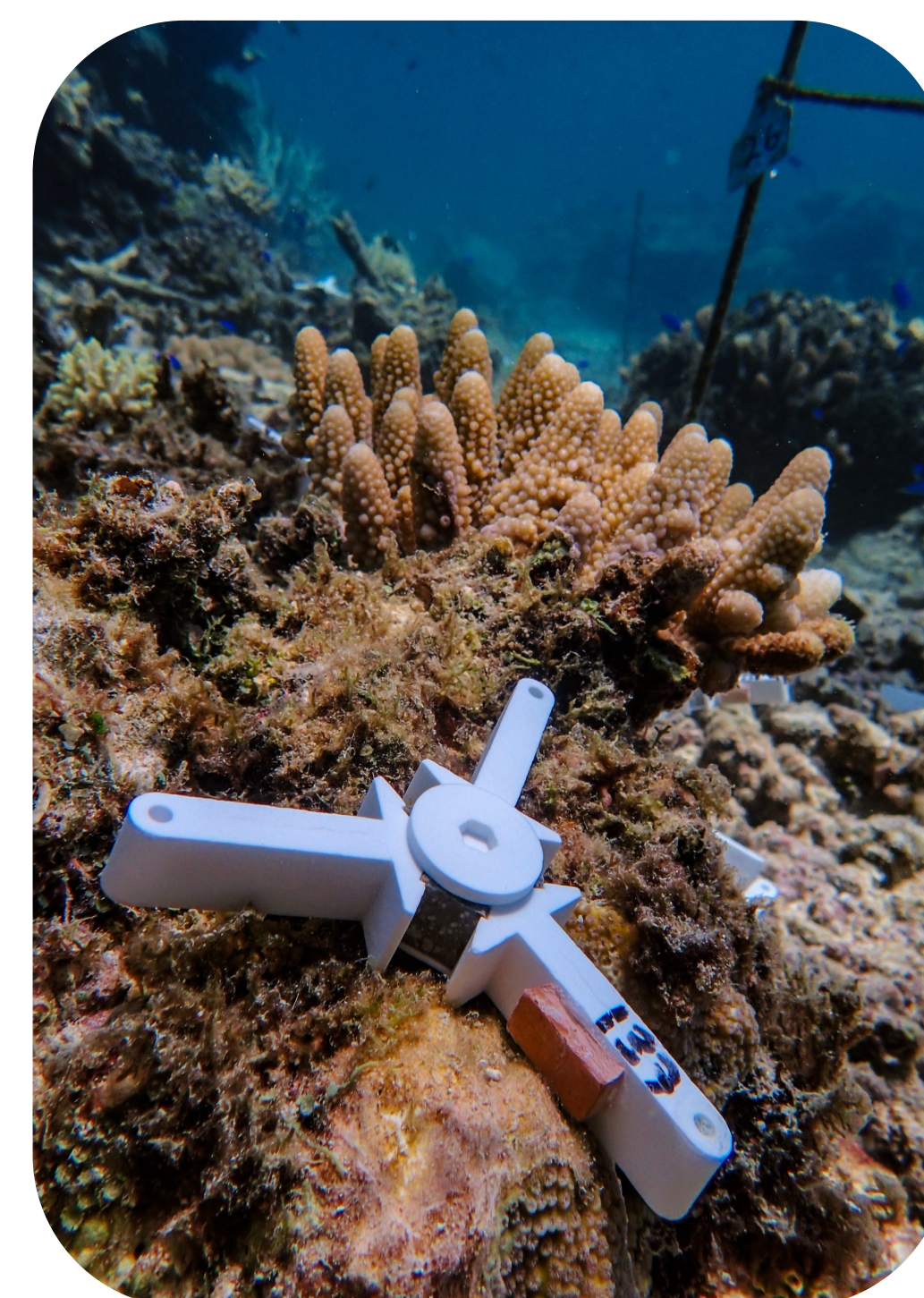
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The Problem

- Coral reefs face critical decline:** climate change and other stressors threaten 70–90% loss of coral species within the next decade, making large-scale restoration urgent
- Coral reseeded offers a promising solution:** by breeding coral spawn, growing them to a resilient stage in aquaculture, and deploying them in ceramic protective devices, reseeded enhances genetic diversity and reef resilience
- Automation is key:** reseeded efforts rely on manual diver labor, limiting scalability; as restoration targets grow to tens of thousands of corals, automated deployment become essential

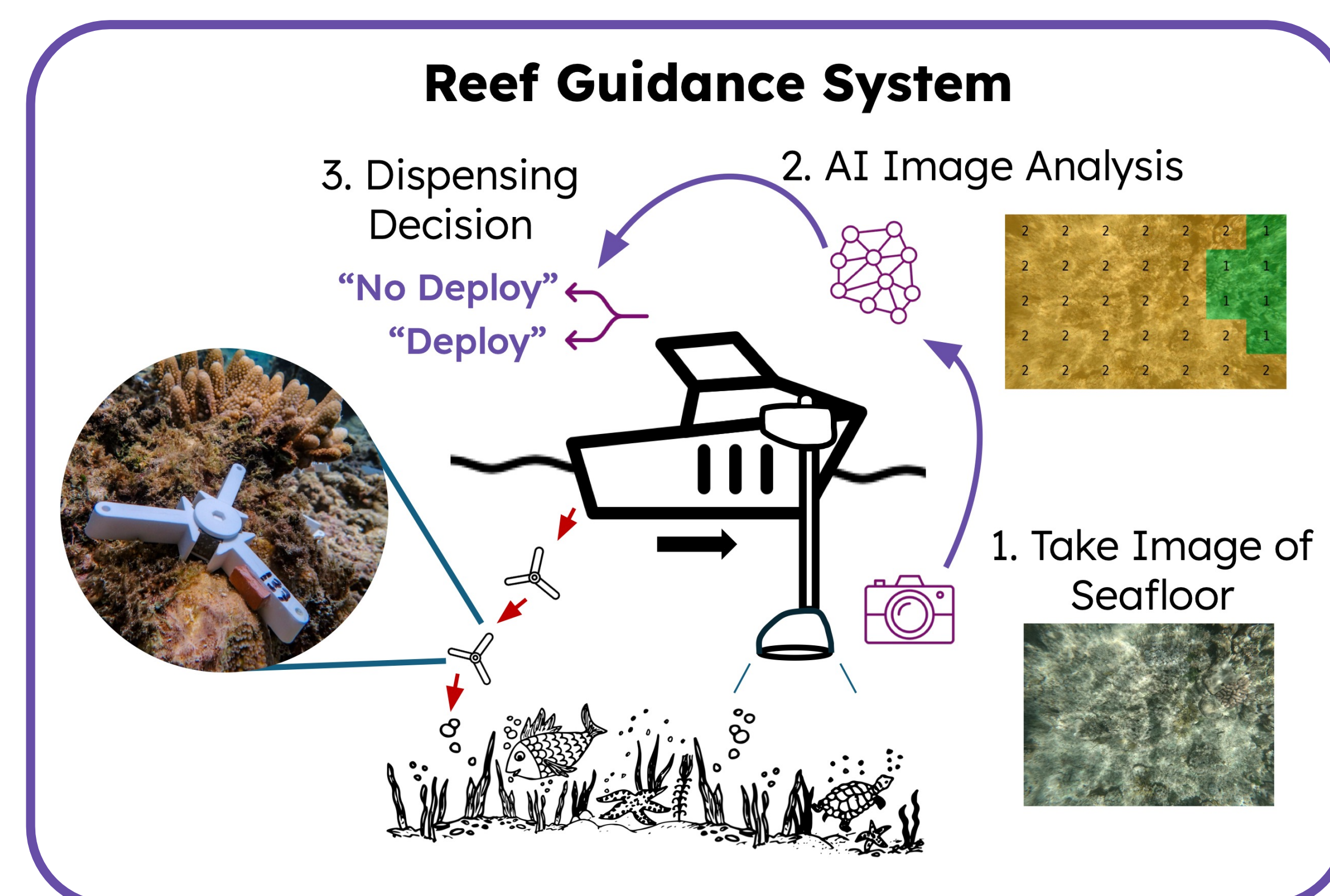


Reef Guidance System (RGS)

First pipeline for broad-scale automated coral reseeded device deployment on the Great Barrier Reef

Reef Guidance System classifies reef substrate in **real-time** to identify suitable deployment locations at depths up to 10m

Runs on an **edge camera unit** aboard the vessel, triggering coral device release automatically



Flexible Pipeline

Components can be selected to suit operational requirements of the reef restoration program (accuracy, annotator availability, inference speed, or hardware setup):

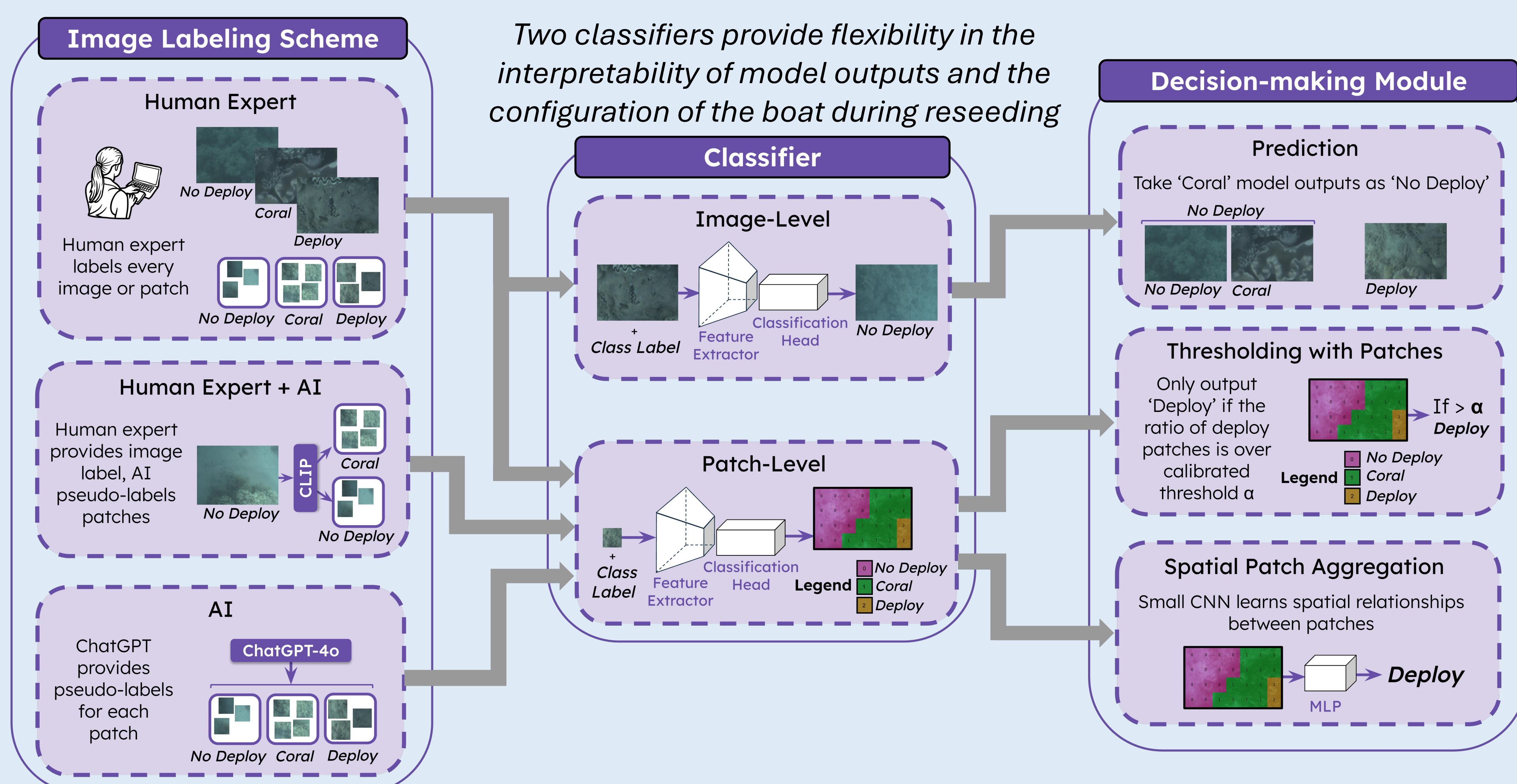
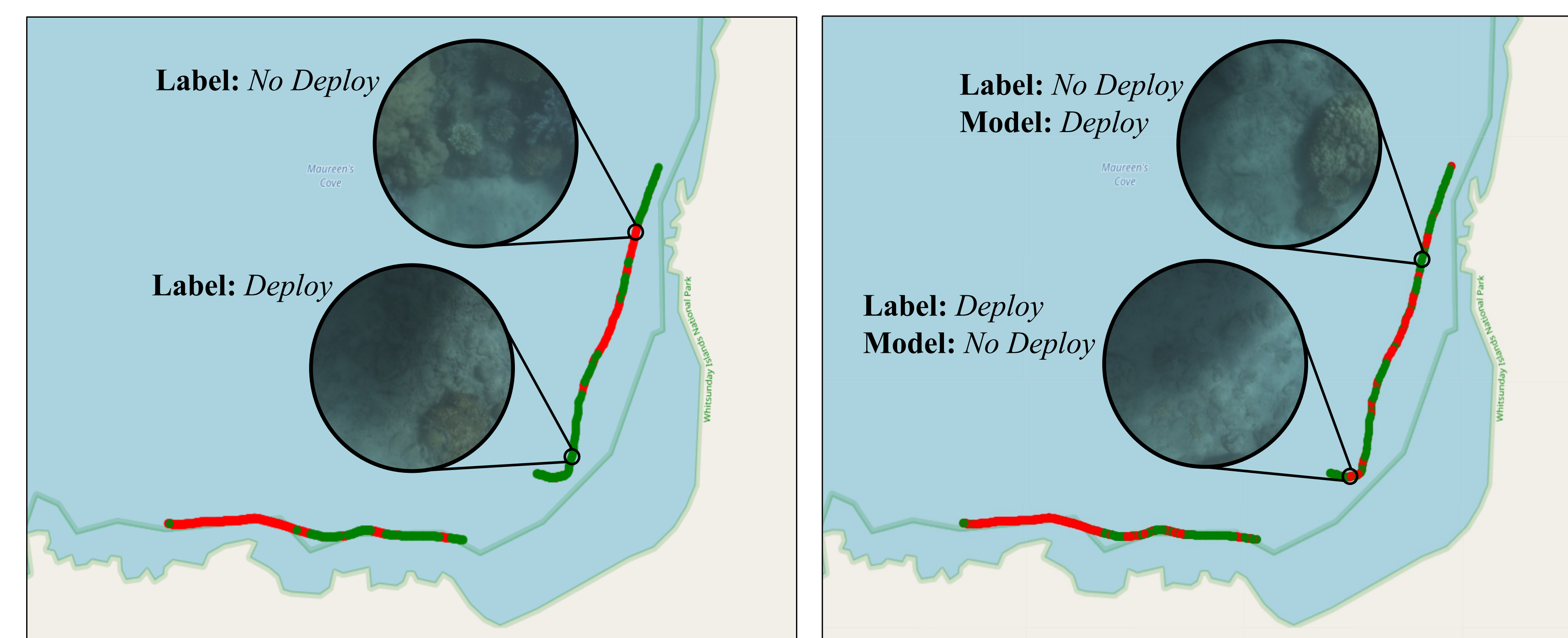


Image labeling schemes offer differing levels of human expert input: this should be selected based on the availability of experts to annotate data of new locations and environmental characteristics

We present three alternatives for dispensing decision-making. These differ in accuracy and inference speed

Results

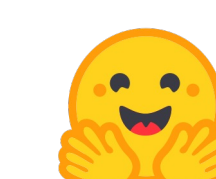
We validate the RGS at five sites across the Great Barrier Reef and find that the accuracy of our AI-driven deployment system is comparable to an expert marine ecologist, while overcoming human limits and significantly increasing the potential for broad-scale coral reseeded



(a) Expert Marine Scientist Decisions

(b) Our RGS Pipeline Decisions

RGS achieves 77.8% deployment accuracy, 89.1% accuracy for sub-image patch classification, and real-time model inference at **5.5 frames per second** on an NVIDIA Jetson Orin



Our data is available on HuggingFace:

<https://huggingface.co/datasets/QCR-Underwater-Perception/reef-guidance-system>



Check out our code on GitHub:

<https://github.com/sgraine/reef-guidance-system>

Acknowledgements

S.R., E.O. and T.F. acknowledge continued support from the Queensland University of Technology (QUT) through the Centre for Robotics. T.F. acknowledges funding from an Australian Research Council (ARC) Discovery Early Career Researcher Award (DECRA) Fellowship DE240100149. S.R., E.O., B.M. and T.F. acknowledge support from the Reef Restoration and Adaptation Program (RRAP) which is funded by a partnership between the Australian Government's Reef Trust and the Great Barrier Reef Foundation.